

Element F
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Design Viability:

- For the automated bear feeder, there were some requirements for what the bear launcher needed
 1. The bear feeder needed enough power in order to be able to launch over the gap between the bear feeder and the bear exhibit
 2. The feeder must be controlled in the sense that it must be able to shoot to different places around the exhibit so that the bears have to gain some sense of foraging for food
 3. The feeder must have the ability to launch food at varying times throughout the day
 4. Additionally, the bear feeder should ideally be able to shoot the bears food autonomously

- Regarding our prototype, we noticed we had the following priorities covered
 1. As our code worked, we were able to obtain the code of the time at any moment in the day, and thus we can shoot the launcher at random times
 2. Working in conjunction with the build team, we have gotten tests in which the launcher shoots at a distance that is more than enough to get the food into the bear exhibit

3. Additionally, we have displayed (as mentioned in previous elements) that the motors we are using with the arduino are able to be spun, as tested with a servo tester, and this means that we can rotate a spinning base beneath the launcher and thus randomize the launch
4. The autonomous launching is the next step within our shooting and we have been searching for ways to keep our arduino running at all times without a computer.
 - a. We note that using a small 9 volt clock battery will work for keeping the rtc on the arduino running at all times.



- a. The following image shows the type of battery that will be used in the RTC and a typical 12 volt 2 amp battery will be used for the arduino, in the best case.
 - b. The type of battery (bottom right) is the ideal battery that is used



